

OWNER PICKUP CHECKLIST

Anticipy Unit 001

Starting point: XIAO boards only. Everything else below must be bought or printed.

Target	Build path	Deliverable
Wednesday afternoon	Local parts + black PETG	1 closed, battery-powered investor unit
After Unit 001 passes	Freeze exact lots/files	9 more units + 2 unassembled spare kits

ONLY RED ITEM No local store has yet confirmed the exact Renata battery on its shelf. Call first. Do not buy a similar-looking pouch cell.

Your shopping trip - do this in order

1. Get the exact battery first

Call all three stores before driving. Read this sentence word-for-word:

PHONE SCRIPT "I need exact Renata ICP501022UPM, Renata part 100640. It must have the manufacturer label and matching datasheet/UN38.3 record. Do you physically have it in your hand? Please hold one for Omar and do not substitute anything."

Call	Store	Address / hours	What counts as YES
1	Cadex Electronics 604-231-7777	22000 Fraserwood Way, Richmond 08:00-16:30	Staff physically holds exact 100640 and can show label/docs
2	Swiss Watch Parts 604-685-9949	Suite 626, 602 W Hastings, Vancouver 09:00-16:00	Staff physically holds exact 100640 and can show label/docs
3	Vancouver Battery 604-737-8463	1875 Ontario St, Vancouver 09:00-17:00	Staff physically holds exact 100640 and can show label/docs

- [] Buy exactly 1 battery only after the label says ICP501022UPM / 100640.
- [] Ask the supplier to hold 11 more from the same lot, unpaid, until Unit 001 passes.
- [] Photograph both sides, the label, polarity and lot code before leaving.
- [] If every store says NO, buy no battery. A donor, drone, vape or generic pouch cell is an automatic reject.

2. Pick up the complete local electronics bag

Go to: Lee's Electronic Components, 4131 Fraser St, Vancouver | 604-875-1993 | 09:00-17:30

Tell the counter: "Please physically count these exact PIDs. No substitutions."

Buy	Lee PID	Exact item	Why
1	160266	Matter3D black PETG, 1.75 mm, 1 kg	Print full shell and fixtures
12	10431	Sealed 3 V coin motor, listed 10 x 2.7 mm	Haptic; one used, rest batch/spares
12	170433	NTR4101P P-channel MOSFET, SOT-23	Active-low motor switch
3 packs	20011	BAS16 SOT-23 diode, 5 per pack	Motor flyback protection
1 pack	17426	10 kohm 0603 resistor, 20 per pack	Gate pull-up
3 packs	6836	M2 x 10 flat countersunk screws, 10 per pack	Two close each unit; inspect 90-degree head

Known listed subtotal: C\$108.55 before tax. Battery and generic bench consumables are extra.

- [] Before leaving Lee's, count every bag and check each PID against the table.
- [] Ask the hardware guy once: "Are you bringing fine wire, solder/flux, ≤ 0.05 mm insulation film, thin electronics-safe adhesive and acoustic mesh?" If YES, buy none. If NO, buy enough for one unit at Lee's and label the bag BENCH CONSUMABLES.

3. Bring the builder exactly three labelled bags

- [] BATTERY: one exact Renata 100640 plus its label/documents.
- [] ELECTRONICS: motor, MOSFET, diode and resistor bags.
- [] MECHANICAL: black PETG and M2 screws.

Do not spend money on these for Wednesday

- microSD cards or an SD daughterboard
- custom PCB assembly
- aluminum machining, Xometry or JLC enclosures
- a random battery, donor cell or Lee PID8834
- an exposed-rotor barrel motor

HARDWARE BUILDER BRIEF

Build Anticipy Unit 001

Owner hands you XIAO boards and the three labelled pickup bags. Follow this brief in order; do not improvise substitutions.

DELIVERABLE One clean black PETG pendant, nominally 51 x 22 x 14.5 mm, with live BLE audio to the Anticipy iPhone app, battery operation/charging and pendant haptic. No microSD and no offline audio backlog.

A. Freeze the build path

- [] Board is physically marked XIAO nRF52840 Sense and has no pin headers.
- [] Battery is exact Renata ICP501022UPM / part 100640, protected, <=24 x 10 x 5.5 mm finished.
- [] Motor body passes the supplied 10.2 x 10.2 x 2.7 mm maximum gauge.
- [] Use only the Lee NTR4101P high-side circuit and the matching ACTIVE-LOW UF2 in this brief.
- [] Write board ID, battery lot, motor bag/PID, filament lot and builder name on the release record.

B. Print queue - start this first

Qty	Print this exact STL	Use
2	PRINT_FULL_SHELL_body_exact_historical_geometry.stl	One sacrificial body + one final body
2	PRINT_FULL_SHELL_hooked_face_exact_historical_geometry.stl	One sacrificial face + one final face
1 each	PRIMARY_RENATA_ONLY_M2_screw_retained_midband_rise_4p50mm_nomin al.stl ..._loose_tongue.stl	Choose the one that seats with fingertip pressure
2 prints	PRINT_TWO_PRIMARY_RENATA_ONLY_M2_keyed_retainer_posts_HEAD_DO WN.stl	Sacrificial thread test + final posts + spares
1	PRIMARY_RENATA_ONLY_M2_body_drill_marking_jig_USB_NOTCH.stl	Mark the two closure holes
1	PRINT_DECK_DOWN_then_FLIP_battery_safety_bridge.stl	Keeps XIAO off the battery pouch
1 each	BENCH_ONLY_xiao_plan_fit_gauge.stl ALT_BENCH_ONLY_lower_fit_gauge_batt24_Vytronics10p2.stl	Board/lower-stack dry-fit gauges
1 each	ALT_LOCAL_MEASURE_MAX_battery_24x10x5p5.stl PRIMARY_RENATA_ONLY_MEASURE_MAX_coin_motor_10p2x2p7.stl MEASURE_MAX_finished_driver_island_8x4x2.stl	Reject oversize finished parts

All files are under 01_CAD/PRINT_THESE/ in the supplied build packet.

Printer settings

- Black PETG/PETG-HF; 0.4 mm nozzle; dry filament and record the lot.
- 0.12 mm layers for body, face, band and jigs; 0.16 mm for bridge and gauges.
- Five walls, six top/bottom layers, 40% gyroid. Posts: 100% infill.
- Arachne/thin-wall detection ON. Add a 3 mm brim to each band; seam at USB end.
- Body: flat exterior back on plate, cavity open upward. Face: broad exterior face on plate, hooks upward.
- Inspect slicer preview. Paint support only under true horizontal ledges; block support from hook/slide, USB, microphone and screw interfaces.
- Cool completely on the plate. Reject warp, missing hook walls, stringing in post pilots or visible exterior defects.

C. Prove the empty enclosure before electronics

1. Use the first printed body/face pair as sacrificial. Dry-fit the face and both band variants. Keep only the band that seats with fingertip pressure and lets the face drop and slide freely.
2. Fit the USB-notched body jig on the empty sacrificial body. Transfer both 1.0 mm centres, remove the jig, drill 1.0 mm pilots, then enlarge to 2.2 mm preferred or 3/32 inch allowed. Never exceed 2.4 mm.
3. From outside, make a shallow 90-degree countersink. Test the actual M2 screw repeatedly and stop the instant the head is flush.
4. Thread one sacrificial printed post with the actual M2 screw. Reject splitting, whitening, wobble, stripping or screw bottoming.
5. Assemble the empty body + selected band + posts + face. Run 20 open/close cycles, 100 firm shakes, moderate twist and six-direction 1 m drop onto plywood over concrete.
6. If anything cracks, loosens, rattles, binds or needs force: HOLD. Adjust print/process, print another sacrificial set and repeat from the start.

D. Build the Lee active-low haptic circuit

DO NOT MIX This wiring uses the local NTR4101P P-channel high-side circuit and only the ACTIVE-LOW UF2. The DigiKey active-high UF2 can command the opposite electrical state.

From	To
NTR4101P source	XIAO 3V3
NTR4101P drain	Motor positive
Motor negative	XIAO GND
XIAO D0	NTR4101P gate
10 kohm pull-up	NTR4101P gate to XIAO 3V3
BAS16 cathode	Motor positive / NTR4101P drain
BAS16 anode	Motor negative / GND

- [] Confirm the actual NTR4101P and BAS16 SOT-23 pinouts from the received markings/datasheets before soldering.
- [] Build, clean, strain-relieve and insulate the complete driver island. It must pass MEASURE_MAX_finished_driver_island_8x4x2.stl without friction.
- [] Never power the motor directly from GPIO. Prove motor OFF during boot, reset, disconnect, system-off and watchdog recovery before battery power.

E. Flash and commission the board

Use only: 02_FIRMWARE/LOCAL_NTR4101P_FALLBACK_DO_NOT_MIX/

UF2: HOLD_FOR_PHYSICAL_RELEASE_TEST_Anticipy_0.9.3_owner2_live_50mA_LOCAL_NTR4101P_ACTIVE_LOW.uf2

SHA-256: 54b12eefb226d886ae374043ad2e5e409f5301956940de8bb13f76d084d14299

7. Keep the battery disconnected. Perform a controlled full internal erase; sanitize external QSPI on a reused board; restore the compatible bootloader/layout and this UF2.
8. Power by USB in a private room. A zero-bond boot opens a 120-second commissioning window. Pair the intended iPhone immediately.
9. Wait more than two seconds for settings to store. Power-cycle. Prove Phone A reconnects and Phone B cannot connect, read audio or trigger haptic.
10. On USB/bench power, prove intelligible live audio, short/medium/long haptic, reconnect, disconnect reporting, motor-off behavior and no reset/heat.

F. Battery gate - before connecting it

- [] Record maker, exact MPN, lot/date code, protection statement, datasheet and matching UN38.3 summary.
- [] Measure the finished pack including PCM, wrap, seams and folded tabs. It must pass the 24 x 10 x 5.5 mm gauge without friction.
- [] Meter polarity. Never trust wire colour. Never solder to, fold, sand, rewrap or press the pouch.
- [] Use only the factory leads. Leave a relaxed bend and strain relief; keep every hard part and solder joint off both broad pouch faces and all edges.
- [] The application target is about 50 mA charge. Measured worst-case battery-branch charge current must remain below the pack's 80 mA maximum with margin.

G. Stack and close the final unit

11. Use the second, cosmetically clean body/face pair. Deburr and vacuum it. No plastic or metal swarf may remain.
12. Apply no more than 0.05 mm total electrical floor insulation.
13. Use the supplied lower-layout gauge to position the exact Renata battery and gauged coin motor. Bond the motor only to rigid shell structure, never to the cell.
14. Install the insulated driver island in its modeled 8 x 4 x 2 mm zone. Keep solder, wire bundles and metal-backed adhesive away from the XIAO antenna.
15. Seat the printed safety bridge on rigid shelf structure. Put the XIAO above the battery without loading it; USB at the shell opening and microphone at the straight gasketed acoustic path.
16. With the cell still disconnected, perform continuity, polarity and current-limited checks. Then connect the qualified cell through an inline meter.
17. Place 0.05 mm witness film over the pouch and every high point. Close with fingertip pressure, reopen and reject any ridge, mark, pinched wire or shifted part.
18. Install keyed posts and M2 screws. Alternate half-turns only until the band cannot lift. Screws must be flush, not bottomed, and have no inward sharp end.

H. Release test - every box must pass

- [] Cold battery boot 10/10; no motor pulse during boot/reset/disconnect/system-off/watchdog.
- [] Measured combined board + motor current ≤ 160 mA peak and ≤ 80 mA continuous; no reset or heat.

- [] Phone A receives intelligible closed-case live audio for 60 minutes and reconnects 20/20; Phone B is rejected.
- [] Short/medium/long haptic 25 times each with no microphone buzz, RF loss, movement or abnormal temperature.
- [] Closed-case BLE passes at 5 m and 10 m, including intended body-worn orientation.
- [] Charge from 20-30% to at least 80% with measured current inside the exact pack limit; no pouch change or abnormal heat.
- [] Two-hour combined audio/BLE/haptic soak passes; shell/cell rise stays below 10 C over ambient.
- [] Six-direction 1 m drop, 100 firm shakes, moderate twist and normal USB side-load cause no opening, crack, rattle or reset.
- [] Reopen after abuse: pouch has no mark, dent, crease, rub or lead strain. Reclose and repeat a 15-minute audio/reconnect/haptic test.
- [] Exterior is clean and product-like: aligned seam, flush screws, clean USB/mic openings and no warp, stringing, tool marks or raw holes.

RELEASE RULE One failed box means HOLD, repair and full retest. Do not hide a missing function, substitute a part, tighten harder against a fit problem or ship an untested battery unit.

I. After Unit 001 passes

19. Freeze the exact battery lot, motor bag/identity, driver parts, firmware hash, shell files, band variant, filament/profile and app version.
20. Only then buy the held 11 same-lot batteries. Assemble Units 002-010; keep two labelled unassembled spare kits.
21. Repeat the per-unit identity, fit, current, charge, audio, haptic, RF, thermal, abuse and pouch-inspection record. Any substitution returns one unit through the full Unit 001 gate.

Builder release record

Field	Record
Unit serial	_____
Builder / date	_____
Battery MPN / lot	_____
Motor / driver identity	_____
Firmware SHA verified	YES / NO
Peak / continuous current	_____ mA / _____ mA
Two-hour soak	PASS / HOLD
Final decision	PASS / HOLD

Release description: functional production-intent investor sample. Not yet certified, waterproof, production-validated or customer-sale-ready.